PRODUCT USER MANUAL

For
OMI_CIRCULATION_BOUNDARY_ATLANTIC_gulf_stream_destabilization_point

Issue: 1.0

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Approval Date : MARCH 2026

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RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
1.0	2025/11/12	All	Initial version	K Grassi	M. I. Pujol

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GLOSSARY AND ABBREVIATIONS

MOI	Mercator Ocean International
NetCDF	Network Common Data Form
OMI	Ocean Monitoring Indicator
PUM	Product User Manual
QUID	Quality Information Document

DATA ACCESS

After registration, you will be able to downland our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

1) INTRODUCTION

a) Summary

This document is the user manual for the CMEMS Ocean Monitoring Indicator (OMI) OMI_CIRCULATION_BOUNDARY_ATLANTIC_gulf_stream_destabilization_point. It contains 1 OMIs :

- OMI_CIRCULATION_BOUNDARY_ATLANTIC_gulf_stream_destabilization_point

The OMI provides a time series of the destabilisation point index from 1993 until the present.

The product is published on the CMEMS dissemination server after automatic and delayed-mode quality controls. The product is available online and disseminated through the CMEMS Information System. Files downloaded are in NetCDF format and follow CF-1.7 convention.

Disclaimer: The quality of the product may vary during the proposed time series depending on the possible update of the system.

The Gulf Stream destabilization point product is described in the Quality Information Document (QUID) given in section V. See also News flash.

2) DESCRIPTION OF THE PRODUCT SPECIFICATION

a) General Information

Product Lines	OMI_CIRCULATION_BOUNDARY_ATLANTIC_gulf_stream_destabilization_point
Geographical coverage	WEST: 80°, East: 50°, North: 40° et South 45
Variables	Time Latitude: latitude of the destabilization point Longitude: longitude of the destabilization point ci_latitude: confidence interval of the latitude of the destabilization point ci_longitude: confidence interval of the longitude of the destabilization point
Available time series	[1993, -1Y]
Temporal resolution	yearly
Horizontal resolution	/
Integration depth (for example)	Sea Surface
Format	NetCDF4

- All data is in NetCDF4 CF1.7 format.
- Refer to the "Data Access" section for information on data access mechanisms.

b) Production System Description

The estimates are derived from the daily maps of both ADT from satellite altimetry (SEALEVEL_GLO_PHY_L4_MY_008_047) were averaged to produce monthly maps (see QUID for details of each product)

c) Dataset details

omi_circulation_boundary_atlantic_gulf_stream_destabilization_point is defined as the first longitudinal position where the **latitudinal variance** of the current reaches **0.42°²**, which represents half of the maximum variance observed across the mean paths from 1993 to 2022. This threshold marks the onset of Gulf Stream meanders, and its position is calculated using the **25 cm sea surface height (SSH) contour**, selecting the **northernmost latitude** within each **0.5° longitudinal bin**. To smooth interannual fluctuations and reduce the influence of high-frequency variability, a **5-year running mean** is applied to the time series of the destabilization point.
Time: units: "days since 1950-01-01" Calendar: "proleptic_gregorian"Time, gregorian calendar longitude: long_name: longitude of the destabilization point standard_name: longitude_of_the_destabilization_point units: degrees_east latitude: long_name: latitude of the destabilization point standard_name: latitude_of_the_destabilization_point units: degrees_north ci_latitude: long_name: 95% confidence interval of the latitude of the destabilization point standard_name: confidence_interval_latitude_of_the_destabilization_point ci_longitude: long_name: 95% confidence interval of the longitude of the destabilization point standard_name: confidence_interval_longitude_of_the_destabilization_point

3) FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4) FILES NOMENCLATURE

OMI files nomenclature when downloaded through the CMEMS FTP Services is based on the following nomenclature:
<datasetname>_<timestep>_P<productiontime>.nc
Where :

- **<datasetname>** is the dataset reference .
- **<timestep>** is the first time step contained in the file.
- **<productiontime>** is the production time when the file was generated.
- **.nc** is the standard NetCDF filename extension.

Example:
OMI_CIRCULATION_BOUNDARY_ATLANTIC_gulf_stream_destabilization_point_1993_P20250331.nc

An exhaustive description of the files nomenclature when downloaded through the web portal is available here : Files Nomenclature and Format

More information on product and dataset nomenclatures is available here Products and Datasets nomenclature

5) REFERENCES

Quality Information Document

https://documentation.marine.copernicus.eu/QUID/CMEMS-SL-QUID-CIRCULATION-BOUNDARY-gulf-stream-destabilization-point.pdf

https://documentation.marine.copernicus.eu/QUID/CMEMS-MOB-QUID-HEALTH-global-carbon-ph-area-averaged.pdf
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